

The Financial–Epistemic Classification of Political Parties

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This paper develops a financial–epistemic framework for the classification of political parties. Rather than classifying parties exclusively by ideology, electoral performance, or constitutional role, the proposed framework studies political organisations through two complementary representations.

The first representation is financial and identifies parties by their ownership configurations within a monetary architecture. The second representation is epistemic and identifies parties by their nation, political position, and mode of political knowledge transmission.

The paper proposes a mathematical correspondence between these representations under explicit modelling assumptions. The resulting framework combines combinatorics, graph theory, institutional economics, political economy, and computational classification.

The mathematical development builds upon the ownership architecture introduced in the previous paper $P_{\max} \leq 60$, which established an upper bound on the number of admissible political parties under a four-nation, five-currency institutional model.

The paper ends with “The End”

Contents

I	2
1 Introduction	3
2 Relationship with the Previous Paper	3
3 Motivation	4
4 Financial Classification	4
5 Epistemic Classification	4
5.1 Level I	4
5.2 Level II	5
5.3 Level III	5
6 Cardinality of the Epistemic Space	5
7 Research Objective	6
8 Outline of the Paper	6

II	6
9 The Financial–Epistemic Representation	6
10 Classification Spaces	6
11 Existence of Bijections	7
12 Financial Feature Maps	8
13 Epistemic Feature Maps	8
14 Canonical Correspondence	8
15 Graph Representation	9
16 Representation Principle	10
III	10
17 Institutional Interpretation	10
18 Political Economy	10
19 Financial Interpretation	11
20 Graph-Theoretic Interpretation	11
21 Statistical Formulation	11
22 Entropy	12
23 Algorithmic Classification	12
24 Computational Complexity	13
25 Data Science	13
26 Artificial Intelligence	14
27 Discussion	14
IV	14
28 Generalisations	14
28.1 Alternative Epistemic Structures	14
29 Limitations	15
30 Future Research	15
31 Summary	15
32 Concluding Remarks	16

Part I

1 Introduction

Political parties have traditionally been classified according to ideology, constitutional status, historical origin, or electoral behaviour.

Examples include

- conservative,
- liberal,
- socialist,
- social democratic,
- nationalist,
- libertarian,
- religious,
- environmental,
- regional.

Such classifications provide valuable descriptive information but are not generally derived from an underlying mathematical model.

The objective of the present paper is different.

Instead of beginning with ideology, we begin with institutional structure.

Political parties are regarded as organisations possessing financial characteristics, institutional relationships, and mechanisms through which political knowledge is transmitted.

This perspective naturally produces two distinct mathematical representations.

2 Relationship with the Previous Paper

The previous paper entitled $P_{\max} \leq 60$ introduced an ownership-based definition of political parties.

Within that framework, a political party is characterised by

1. one domestic currency,
2. ownership of a non-empty subset of central banks,
3. ownership of the international market portfolio.

Under the modelling assumptions of that paper, the ownership space contains at most

60

admissible ownership configurations.

The present paper investigates whether those ownership configurations admit an alternative but equivalent classification.

3 Motivation

Classification occupies a central role throughout mathematics, economics, statistics, computer science, and political science.

A useful classification should satisfy three desirable properties.

1. Completeness.
2. Internal consistency.
3. Computational tractability.

The ownership classification satisfies these properties within the institutional architecture.

However, it is economically useful to interpret ownership structures in terms of political organisation.

The present paper therefore introduces an epistemic classification.

4 Financial Classification

Definition 1. The financial classification of a political party is the ordered triple

$$\Pi = (c, S, M),$$

where

- c denotes the domestic currency,
- S denotes the non-empty subset of central banks owned by the party,
- M denotes ownership of the international market portfolio.

The financial representation describes the institutional balance sheet associated with each political organisation.

5 Epistemic Classification

Political organisations also differ in the manner in which political knowledge is created, transmitted, and preserved.

The present framework introduces three hierarchical levels.

5.1 Level I

The first level identifies the sovereign nation.

Assume

$$N_{\max} = 4.$$

The available nations are

$$\{N_1, N_2, N_3, N_4\}.$$

5.2 Level II

Each nation possesses five political positions.

Position	Description
1	Far Left
2	Left
3	Center
4	Right
5	Far Right

5.3 Level III

Political knowledge is transmitted through one of three mechanisms.

Code	Transmission Mode
1	Public Broadcast
2	Private Teaching
3	Protected Learning

Definition 2. The epistemic classification of a political party is the ordered triple

$$\Lambda = (N, L, T),$$

where

- N denotes the nation,
- L denotes the political position,
- T denotes the knowledge-transmission mode.

6 Cardinality of the Epistemic Space

The epistemic classification contains

- four nations,
- five political positions,
- three transmission mechanisms.

Consequently,

$$|\Lambda| = 4 \times 5 \times 3 = 60.$$

The epistemic space therefore contains sixty admissible classifications.

7 Research Objective

The equality

$$|\Pi| = |\Lambda| = 60$$

suggests the possibility of a structural correspondence between financial ownership configurations and epistemic classifications.

The principal objective of the present paper is to determine conditions under which there exists a mapping

$$\Phi : \Pi \rightarrow \Lambda,$$

and to investigate when this mapping may be chosen to be a bijection.

The existence of such a mapping would permit every admissible financial ownership configuration to be represented uniquely by an epistemic classification.

8 Outline of the Paper

The remaining sections develop the theory in stages.

Part II constructs the mathematical representation theorem, establishes sufficient conditions for the existence of financial–epistemic correspondences, and develops the graph-theoretic formulation.

Subsequent parts investigate computational, institutional, economic, and statistical interpretations of the resulting classification framework.

Part II

9 The Financial–Epistemic Representation

The preceding section introduced two finite classification spaces.

The financial classification space is

$$\Pi = \{(c, S, M)\},$$

while the epistemic classification space is

$$\Lambda = \{(N, L, T)\}.$$

Both spaces contain sixty admissible elements under the modelling assumptions.

The present section studies mappings between these two representations.

10 Classification Spaces

Definition 3. The financial state space is

$$\mathcal{F} = \{(c, S, M) : c \in \mathcal{C}, S \subseteq \mathcal{B}, S \neq \emptyset, M = 1\}.$$

Since

$$|\mathcal{C}| = 4,$$

and

$$|\mathcal{B}| = 4,$$

it follows that

$$|\mathcal{F}| = 4(2^4 - 1) = 60.$$

Definition 4. The epistemic state space is

$$\mathcal{E} = \{(N, L, T)\},$$

where

$$N \in \{N_1, N_2, N_3, N_4\},$$

$$L \in \{FL, L, C, R, FR\},$$

and

$$T \in \{PB, PT, PL\}.$$

Therefore,

$$|\mathcal{E}| = 4 \times 5 \times 3 = 60.$$

11 Existence of Bijections

The first mathematical observation follows immediately from finite set theory.

Theorem 1. *Assume*

$$|\mathcal{F}| = |\mathcal{E}| = 60.$$

Then there exists at least one bijection

$$\Phi : \mathcal{F} \rightarrow \mathcal{E}.$$

Proof. Finite sets having equal cardinality are equipotent.

Consequently, there exists a one-to-one and onto correspondence between their elements.

Therefore,

$$\Phi : \mathcal{F} \rightarrow \mathcal{E}$$

exists.

□

Remark 1. The theorem establishes existence only.

It does not identify a preferred correspondence.

Indeed,

$$60!$$

distinct bijections exist between the two finite sets.

Therefore, additional economic structure is required to identify a canonical classification.

12 Financial Feature Maps

The ownership configuration contains considerably more information than its combinatorial encoding.

Each ownership configuration possesses measurable financial characteristics.

Definition 5. A financial feature map is a function

$$F : \mathcal{F} \rightarrow \mathbb{R}^k,$$

assigning a numerical feature vector to every ownership configuration.

Examples include

- domestic monetary exposure,
- number of central banks owned,
- ownership concentration,
- diversification,
- liquidity,
- systemic interconnectedness,
- governance influence.

13 Epistemic Feature Maps

Similarly, each epistemic classification possesses observable characteristics.

Definition 6. An epistemic feature map is

$$G : \mathcal{E} \rightarrow \mathbb{R}^k.$$

Typical coordinates may encode

- political position,
- communication intensity,
- institutional openness,
- organisational centralisation,
- public accessibility.

14 Canonical Correspondence

The preceding constructions motivate a stronger notion of equivalence.

Definition 7. A bijection

$$\Phi : \mathcal{F} \rightarrow \mathcal{E}$$

is called financially consistent if

$$F(x) = G(\Phi(x))$$

for every

$$x \in \mathcal{F}.$$

Thus, the epistemic classification preserves the chosen financial features.

Proposition 1. *Suppose*

1. *F is injective,*
2. *G is injective,*
3. *the images satisfy*

$$F(\mathcal{F}) = G(\mathcal{E}).$$

Then there exists a unique financially consistent bijection.

Proof. Injectivity implies that every feature vector determines a unique financial state and a unique epistemic state.

Equality of the feature images therefore induces a unique correspondence preserving feature vectors.

Hence

$$\Phi = G^{-1} \circ F.$$

The mapping is bijective because both component mappings are bijective between their domains and common image.

□

15 Graph Representation

The financial architecture naturally induces a bipartite graph.

Let

$$V_F = \mathcal{F},$$

and

$$V_E = \mathcal{E}.$$

Whenever

$$\Phi(f) = e,$$

an edge joins

$$f \in V_F$$

to

$$e \in V_E.$$

Consequently,

$$G = (V_F, V_E, E)$$

forms a perfect matching whenever

$$\Phi$$

is bijective.

$$\boxed{\mathcal{F}} \xrightarrow{\Phi} \boxed{\mathcal{E}}$$

16 Representation Principle

The mathematical developments of this section establish that the financial and epistemic classifications may be regarded as two representations of the same finite institutional state space, provided a financially consistent bijection exists.

Subsequent sections investigate the computational, economic, institutional, and statistical implications of this representation theorem.

Part III

17 Institutional Interpretation

The financial–epistemic representation provides two equivalent descriptions of an admissible political organisation under the assumptions of the model.

The financial description records ownership relationships within the monetary architecture.

The epistemic description records the institutional environment within which political knowledge is created, transmitted, and preserved.

Consequently, the representation theorem permits either description to be used without changing the underlying institutional state.

18 Political Economy

Political parties perform several institutional functions, including

1. aggregation of preferences,
2. policy formation,
3. electoral competition,
4. governmental participation,
5. communication with citizens.

Within the present framework, these functions are analysed independently of ideological labels.

Instead, institutional ownership determines the financial identity of the organisation, while the epistemic classification specifies its political organisation.

The resulting representation separates institutional structure from ideological content.

19 Financial Interpretation

Each financial configuration

$$(c, S, M)$$

may be interpreted as a constrained institutional portfolio.
The domestic currency determines monetary denomination.
The subset

$$S$$

determines institutional exposure to the monetary authorities.
Ownership of the international market portfolio represents participation in the global financial system.
Accordingly, the representation theorem may be interpreted as translating an institutional balance sheet into an epistemic classification.

20 Graph-Theoretic Interpretation

Let

$$G = (V, E)$$

be the bipartite graph introduced in Part II.
Every financial state

$$f \in \mathcal{F}$$

is connected by exactly one edge to an epistemic state

$$e \in \mathcal{E}$$

whenever a financially consistent bijection exists.
Hence

$$|V| = 120,$$

while

$$|E| = 60.$$

The graph therefore forms a perfect matching between the two classification spaces.

$$\boxed{\mathcal{F}} \xrightarrow{\Phi} \boxed{\mathcal{E}}$$

21 Statistical Formulation

The ownership space

$$\Omega = \mathcal{F} = \{f_1, \dots, f_{60}\}$$

may be regarded as a finite sample space.
Likewise,

$$\mathcal{E} = \{e_1, \dots, e_{60}\}$$

forms an equivalent finite sample space.

Let

$$P(f_i) \geq 0,$$

with

$$\sum_{i=1}^{60} P(f_i) = 1.$$

The representation theorem induces

$$P(e_i) = P(\Phi^{-1}(e_i)).$$

Consequently, probability measures may be transferred between the two representations while preserving total probability.

22 Entropy

Suppose

$$P_1, \dots, P_{60}$$

denote probabilities assigned to the financial states.

The Shannon entropy equals

$$H = - \sum_{i=1}^{60} P_i \log P_i.$$

Since

$$\Phi$$

is bijective,

the entropy computed on

$$\mathcal{F}$$

equals the entropy computed on

$$\mathcal{E}.$$

Thus, the representation preserves information content.

23 Algorithmic Classification

The representation theorem naturally produces a classification algorithm.

Input

Financial configuration

$$(c, S, M).$$

Output

Epistemic configuration

$$(N, L, T).$$

Pseudo-code

Read financial configuration.

Compute feature vector.

Locate matching epistemic feature vector.

Return corresponding
(N,L,T).

24 Computational Complexity

Assume the financially consistent bijection has already been constructed.

If implemented using a lookup table containing the sixty correspondences, classification requires constant time,

$$O(1).$$

If the correspondence is instead computed through a search over all admissible states, the worst-case complexity is

$$O(60),$$

which is still constant with respect to the fixed architecture.

25 Data Science

The finite state space permits several computational analyses.

Examples include

- network visualisation,
- clustering,
- anomaly detection,
- simulation,
- institutional stress testing,
- scenario analysis.

Because the admissible state space is finite, every possible institutional configuration may be enumerated exactly.

26 Artificial Intelligence

The finite classification space also provides a natural state space for computational models.

Possible applications include

- supervised classification,
- graph neural networks,
- reinforcement learning,
- symbolic reasoning,
- knowledge graphs.

These applications treat the mathematical framework as a discrete representation of institutional states rather than as evidence that real political systems necessarily follow the model.

27 Discussion

The financial and epistemic classifications provide two distinct coordinate systems for describing the same institutional object under the assumptions introduced in Parts I and II.

The representation theorem therefore enables researchers to move between financial ownership structures and epistemic classifications without altering the underlying state.

This interpretation connects combinatorics, institutional economics, graph theory, political economy, statistics, and computer science through a common mathematical framework while leaving empirical calibration and validation as subjects for future investigation.

Part IV

28 Generalisations

The financial–epistemic framework admits several mathematical extensions.

28.1 Alternative Epistemic Structures

The present paper adopts

- four nations,
- five political positions,
- three transmission modes.

Other classification hierarchies may be considered.

For example,

- additional political positions,
- multiple communication channels,
- hierarchical educational systems,
- federal institutional structures.

The representation theorem continues to apply whenever the financial and epistemic state spaces possess equal finite cardinality.

29 Limitations

The mathematical results depend upon the assumptions introduced throughout the paper.

In particular,

1. the ownership model,
2. the construction of the epistemic classification,
3. the existence of financial feature maps,
4. the assumptions required for a financially consistent bijection.

The paper establishes structural properties of the proposed model.

It does not establish that actual political parties or political systems necessarily satisfy these assumptions.

Consequently, empirical investigation would be required before applying the framework to observed political organisations.

30 Future Research

Several directions remain open.

1. Empirical estimation of financial feature maps.
2. Construction of canonical financially consistent bijections.
3. Dynamic models of political-party formation.
4. Time-varying institutional ownership.
5. Weighted central-bank ownership.
6. Probabilistic financial classifications.
7. Network dynamics.
8. Comparative institutional analysis across countries.
9. Computational implementations for institutional simulation.

31 Summary

The paper introduced two complementary descriptions of political organisations.

The financial representation

$$(c, S, M)$$

describes institutional ownership.

The epistemic representation

$$(N, L, T)$$

describes nation, political position, and knowledge-transmission mechanism.

Under the assumptions of the model,

$$|\mathcal{F}| = |\mathcal{E}| = 60.$$

Elementary finite-set theory therefore guarantees the existence of bijections between the two representations.

Additional assumptions concerning feature maps permit the construction of financially consistent correspondences that preserve selected institutional characteristics.

Accordingly, the framework provides a mathematical language through which financial ownership and epistemic organisation may be analysed within a unified classification system.

32 Concluding Remarks

The principal contribution of this paper is the introduction of a financial–epistemic perspective on political-party classification.

The mathematical development combines

- finite combinatorics,
- graph theory,
- institutional economics,
- political economy,
- statistical modelling,
- computational classification.

Rather than replacing traditional ideological classifications, the proposed framework provides an alternative structural representation that may be useful for theoretical analysis, simulation, and computational modelling under the stated assumptions.

References

- [1] Soumadeep Ghosh, $P_{\max} \leq 60$, Kolkata, India. 2026.
- [2] Marshall Hall, *Combinatorial Theory*, Second Edition, John Wiley & Sons, 1986.
- [3] Béla Bollobás, *Modern Graph Theory*, Springer, 1998.
- [4] Douglas B. West, *Introduction to Graph Theory*, Second Edition, Prentice Hall, 2001.
- [5] Gérard Debreu, *Theory of Value*, Yale University Press, 1959.
- [6] Andreu Mas-Colell, Michael D. Whinston, Jerry R. Green, *Microeconomic Theory*, Oxford University Press, 1995.
- [7] Kenneth J. Arrow, *Social Choice and Individual Values*, Second Edition, Yale University Press, 1963.
- [8] Harry M. Markowitz, “Portfolio Selection,” *Journal of Finance*, Vol. 7, No. 1, 1952, pp. 77–91.
- [9] William F. Sharpe, “Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk,” *Journal of Finance*, Vol. 19, 1964, pp. 425–442.
- [10] John F. Nash, “Equilibrium Points in n -Person Games,” *Proceedings of the National Academy of Sciences*, Vol. 36, 1950, pp. 48–49.

Glossary

$\mathcal{F}$ Financial state space.

$\mathcal{E}$ Epistemic state space.

(c, S, M) Financial ownership configuration consisting of domestic currency, central-bank ownership subset, and ownership of the international market portfolio.

(N, L, T) Epistemic classification consisting of nation, political position, and knowledge-transmission mode.

F Financial feature map.

G Epistemic feature map.

Φ Mapping from the financial state space to the epistemic state space.

Financially Consistent Bijection A bijection preserving the selected feature vectors through the relationship

$$F(x) = G(\Phi(x)).$$

Representation Theorem The theorem establishing the existence of bijections between the financial and epistemic classification spaces when they have equal finite cardinality.

The End